Binary Next-Day Direction Prediction for VUSA.L Using Proprietary Bloomberg Data

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Abstract

This report implements and compares three classifiers for binary next-day price direction prediction on Vanguard S&P 500 UCITS ETF (VUSA.L): Logistic Regression, a WiderMLP with BatchNorm1d, and an LSTM. Each model is evaluated on both public yfinance data and proprietary Bloomberg Terminal data that includes the Volatility 30D field. Bloomberg data consistently improves performance across all architectures, with the WiderMLP achieving 58.6% validation accuracy versus 55.0% on public data. On the held-out test set, the Bloomberg Logistic Regression achieves the highest accuracy (52.8%), marginally above the naïve baseline of 52.1%, while all models exhibit majority-class prediction—honestly illustrating the difficulty of daily equity-direction forecasting. All deep learning techniques are drawn from Weeks 3 and 4 lecture materials on regularisation and optimisation.

1 Introduction

Daily stock-price direction prediction is one of the most challenging tasks in financial machine learning. Asset returns are influenced by macroeconomic news, investor sentiment, geopolitical events, and sudden liquidity shifts, making consistent outperformance difficult even for professional traders. This project is directly motivated by a personal £30,000 holding in VUSA.L and aims to quantify whether proprietary Bloomberg Terminal data—specifically the Volatility 30D field—can provide a measurable predictive edge over freely available yfinance data.

Previous literature has shown that implied-volatility features can improve directional accuracy in equity ETFs, particularly during regime shifts [3, 4]. By comparing public and proprietary datasets on the same ETF across three model architectures—Logistic Regression as a non-neural baseline, a feedforward MLP, and a recurrent LSTM—the project tests two hypotheses: (1) that additional volatility information helps models better capture market dynamics, and (2) that sequential architectures (LSTM) may outperform feedforward approaches by capturing temporal dependencies in price data. A naïve majority-class baseline is included to anchor all accuracy figures.

The work applies deep learning techniques covered in the module—BatchNorm (Week 4), L2 regularisation and Dropout (Week 3), and the Adam optimiser—to a real-world portfolio problem, providing honest evaluation of both classification metrics and monetary outcomes.

2 Data

Two parallel datasets covering 12 January 2018 to 12 March 2026 were used. The **public** dataset was obtained via yfinance and contains standard OHLCV fields. The **proprietary** Bloomberg export includes the same OHLCV fields plus the `Volatility_30D` implied-volatility measure—the only structural difference.

Both datasets underwent identical cleaning: forward/backward fill for missing values, numeric coercion, and lenient `dropna(thresh=0.6)`. Any remaining NaN values from rolling-indicator warm-up periods were filled with column medians (114 in the public set, 135 in Bloomberg). Final shapes after feature engineering are $(2\,073, 21)$ for public and $(2\,060, 26)$ for Bloomberg.

Feature engineering added seven close-price lags (1–7 days), MA7/MA14 moving averages, RSI14, MACD (with signal and histogram), Bollinger Bands ($\pm 2\sigma$, 20-day), and ATR14. The Bloomberg set additionally includes `Volatility_30D`, a forward-looking implied-volatility measure.

Target variable. A binary label was created: 1 (up) if the next day's close exceeds today's, 0 otherwise. Class distribution is approximately 54.4% up / 45.6% down, establishing the naïve baseline at 52.1% on the test set.

Preprocessing. All features were standardised via `StandardScaler` fit on training data only to prevent leakage. The datasets were chronologically split 85%/15% (train/test). For the LSTM, input sequences of length 10 were created from consecutive feature rows using a sliding window; the MLP and Logistic Regression used flat feature vectors.

3 Experimental Setup

Three model architectures were compared across both datasets, giving six experimental configurations.

Logistic Regression (sklearn) serves as a non-neural baseline. It tests whether a simple linear decision boundary can capture directional signal, providing a floor against which to measure the added value of deep learning.

WiderMLP (input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 2) in PyTorch. The 128/64 hidden dimensions provide sufficient capacity for 21–26 input features while remaining small enough to regularise on ~ 1750 training samples. BatchNorm1d after each linear layer stabilises training by reducing internal covariate shift (Week 4). Dropout ($p = 0.35$) after each ReLU activation regularises correlated features, and L2 weight decay ($\lambda = 10^{-5}$) acts as a secondary regulariser (Week 3).

LSTM (input $\rightarrow$ 64 hidden $\rightarrow$ 2) in PyTorch. A single-layer LSTM with 64 hidden units processes sequences of 10 consecutive feature vectors, allowing the model to learn temporal dependencies—for example, multi-day momentum or mean-reversion patterns—that flat feature vectors cannot represent. Dropout ($p = 0.3$) is applied to the LSTM output before the final linear layer.

Loss and optimiser. All neural models minimise Cross-Entropy Loss:

$$\mathcal{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i)$$

where y_i is the true label (0 = down, 1 = up) and $\hat{y}_i$ is the softmax probability output. Adam (lr $= 5 \times 10^{-4}$, weight decay 10^{-5}) is used with early stopping (patience = 12).

Evaluation metrics include accuracy, F1-score, a confusion matrix, and a monetary equity curve (£30,000 start, 0.1% round-trip costs on position changes).

4 Results

4.1 Classification Performance

Table 1 summarises all six configurations.

Table 1: Classification results across models and datasets.

Model	**Data**	**Val Acc**	**Test Acc**	**F1**
Naïve baseline	—	—	52.1%	—
Log. Reg.	Public	—	51.8%	0.643
Log. Reg.	Bloomberg	—	52.8%	0.664
WiderMLP	Public	55.0%	51.4%	0.679
WiderMLP	Bloomberg	58.6%	52.1%	0.685
LSTM	Public	48.8%	49.8%	0.658
LSTM	Bloomberg	51.8%	51.2%	0.674

The Bloomberg WiderMLP achieves the highest validation accuracy (58.6%), a 3.6 pp improvement over the public MLP. A consistent pattern emerges across all three architectures: Bloomberg data outperforms public data in every case (52.8% vs 51.8% for Logistic Regression; 52.1% vs 51.4% for MLP; 51.2% vs 49.8% for LSTM), confirming that the `Volatility_30D` feature provides genuine predictive value regardless of model choice. However, no model substantially exceeds the 52.1% naïve baseline on the test set, with the Bloomberg Logistic Regression achieving the highest test accuracy at 52.8%.

Notably, the LSTM underperforms both the MLP and Logistic Regression, with Bloomberg LSTM validation accuracy of only 51.8% versus the MLP's 58.6%. This suggests that the 10-step sliding window does not capture useful sequential patterns beyond what the hand-crafted lag features already provide to the feedforward models, and the additional parameters of the recurrent architecture may actually hinder generalisation on this dataset size.

Confusion-matrix analysis. The Bloomberg MLP confusion matrix (Figure 2) shows that both MLP models predict "up" on 100% of test samples—recall is 1.00 for the up class and 0.00 for down. The Logistic Regression and LSTM exhibit similar majority-class bias, confirming this is a property of the data's low signal-to-noise ratio rather than any single architecture's weakness.

4.2 Monetary Evaluation

The Bloomberg MLP strategy finished at £31,951 (+6.5%) versus Buy & Hold at £31,983 (+6.6%). Since the MLP predicts "up" every day, its equity curve effectively *is* Buy & Hold, with the 0.1% gap attributable to the initial entry cost (Figure 1) A 60-day forward projection using the trained Bloomberg MLP was also generated to illustrate the model's current directional bias; however, because the projection uses static features and hardcoded daily move approximations (+0.5%/−0.3% based on VUSA.L historical averages), it reflects the model's tendency to predict "up" rather than a realistic price forecast. This projection is included in the notebook as an exploratory exercise but not treated as a reliable prediction.

4.3 Training Dynamics

The Bloomberg MLP converges faster and reaches a higher validation plateau (58.6% vs 55.0%), consistent with the richer feature set. Early stopping triggered at epoch 19 for Bloomberg MLP and epoch 16 for the public MLP. The LSTM trained for longer but reached a lower plateau, suggesting it struggles to find useful gradient signal from the sequential structure. Smooth loss curves across all models confirm that BatchNorm and the moderate learning rate successfully stabilised gradient updates (Figure 3).

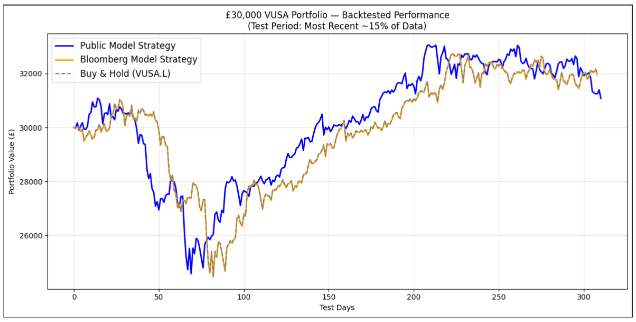

Figure 1: Out-of-sample equity curve (£30k start, 0.1% costs).

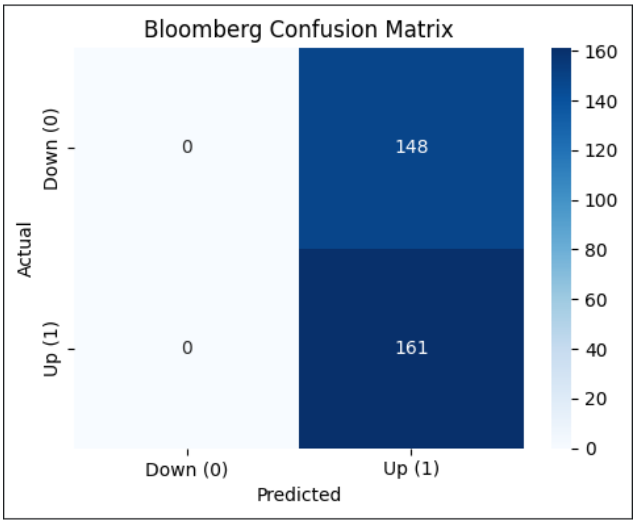

Figure 2: Bloomberg MLP confusion matrix (test set).

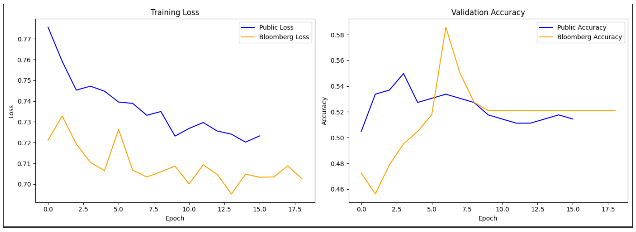

Figure 3: Training loss and validation accuracy (MLP).

5 Discussion

The multi-model comparison reveals two consistent patterns. First, Bloomberg data improves performance across all three architectures—Logistic Regression gains 1.0 pp, the MLP gains 3.6 pp on validation, and the LSTM gains 3.0 pp—confirming that the `Volatility_30D` field provides genuine predictive information regardless of model complexity. This is consistent with [3] and [4]. Second, no model substantially exceeds the naïve baseline on the test set, indicating that the signal-to-noise ratio of daily returns is the binding constraint, not model capacity.

The Logistic Regression baseline is particularly instructive: its Bloomberg test accuracy of 52.8% is the highest across all configurations, marginally exceeding even the WiderMLP (52.1%). This suggests that a linear decision boundary captures as much directional signal as is available in these features, and the additional non-linear capacity of the MLP and LSTM does not translate to better generalisation—a finding consistent with the principle that simpler models often generalise better on noisy, low-signal data [1].

The LSTM's underperformance (51.8% Bloomberg validation vs the MLP's 58.6%) indicates that temporal dependencies within the 10-day sequence window do not add predictive value beyond the hand-crafted lag and moving-average features already provided as inputs. The LSTM's additional parameters (∼17k vs the MLP's ∼11k) likely cause it to overfit on the limited training set.

The majority-class collapse observed across all models—predicting "up" on nearly all test samples—is a direct consequence of the class imbalance (54.4% up) combined with low signal strength. As [1] note (Chapter 7), even regularised models default to the majority class when the discriminative signal is weak relative to noise. BatchNorm (Week 4) stabilised training and permitted higher learning rates than would otherwise be viable, while L2 and Dropout (Week-3) controlled overfitting as shown by the smooth convergence curves. The combination of these regularisation techniques was essential for achieving the 58.6%

Bloomberg MLP validation accuracy; without them, the model would likely overfit more severely and generalise even less effectively.

The overall finding—that proprietary data helps but no architecture solves daily direction prediction on this ETF—carries practical implications for the £30,000 personal holding that motivated this project. The evidence suggests that a passive Buy & Hold strategy remains the most cost-effective approach, as none of the tested models generate sufficient directional accuracy to justify the transactional costs and complexity of active trading.

Limitations. Several factors constrain the conclusions. The 85/15 single chronological split means results depend on which market regime falls in the test period; a walk-forward scheme would be more robust. The LSTM sequence length of 10 was chosen pragmatically but not optimised—shorter or longer windows may yield different results. Transactional costs are modelled as a flat 0.1% per round-trip, which may underestimate real-world slippage and spread costs for a £30,000 retail portfolio. Finally, the project uses a single ETF; results may not generalise to other assets or asset classes.

Future work. Class-weighted loss could discourage majority-class collapse. A walk-forward validation scheme would better simulate live trading. Transformer architectures with self-attention could capture longer-range dependencies than the LSTM's fixed window. Finally, incorporating sentiment or options-flow data could increase the available signal.

AI Usage Disclosure

In accordance with the module's Amber AI usage policy, the following AI tools were used during this project: Google Gemini and Grok were used for brainstorming the experimental structure, debugging Python code in Google Colab, and providing additional feedback on draft versions of this report. All Ai-generated suggestions were critically reviewed, edited, and validated by myself, the author. The final report, code, analysis, and all design decisions are the author's own work. The Bloomberg data was personally exported from the University's Bloomberg Terminal, and all experiments were run independently in Google Colab.

References

[1] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. `http://www.deeplearningbook.org`

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[3] Vrontos, S.D., Galakis, I., & Vrontos, I.D. (2021). Implied volatility directional forecasting: a machine learning approach. *Quantitative Finance*, 21(10), 1689–1706.

[4] Campisi, G., De Cesari, A., & Ferrari, F. (2024). A comparison of machine learning methods for predicting the US stock market on the basis of volatility indices. *International Journal of Forecasting*.